Title: Predict customer purchase decisions for an online retail company

1. Objective

The main objective of this project is to analyze customer behavior and predict whether a customer is likely to purchase a product from an online retail company using machine learning techniques. Purchase prediction helps businesses understand customer preferences, identify potential buyers, improve marketing strategies, and increase sales. It also supports personalized product recommendations and customer engagement.

2. Dataset Description

This project uses an online retail customer dataset containing 2,000 rows and 10 columns. The dataset includes customer age, gender, annual income, time spent on the website, pages viewed, previous purchases, cart items, discount usage, device type, and purchase status. These features help analyze customer behavior and identify factors that influence purchasing decisions. The purchase status column is used as the target variable, where 1 indicates that the customer purchased a product and 0 indicates that the customer did not purchase a product.

3. Approach and Techniques

The project is developed using Python in Google Colab. First, the dataset is created and loaded using the Pandas and NumPy libraries. The data is explored to understand its structure, identify missing values, and examine the distribution of purchase decisions. Categorical variables, such as gender and device type, are converted into numerical values using one-hot encoding. The dataset is then divided into training and testing sets using the train-test split technique. A Random Forest Classifier from the Scikit-learn library is trained to predict customer purchase decisions. Model performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Feature importance is also analyzed to identify the factors that influence customer purchasing behavior.

4. Findings

The project produces predictions indicating whether customers are likely to purchase products and estimates their purchase probabilities. The classification report and confusion matrix help evaluate the model's performance in identifying potential buyers and non-buyers. Feature importance analysis highlights the customer attributes that contribute most to the model's predictions, such as time spent on the website, pages viewed, previous purchases, cart items, and discount usage. Since the dataset is synthetically generated, the results demonstrate the working of the machine learning model rather than actual customer purchasing patterns. The model's performance on real-world data may differ significantly.

5. Conclusion

This project demonstrates the application of machine learning techniques to predict customer purchase decisions in an online retail company. It helps analyze customer behavior, identify potential buyers, and understand the factors associated with purchasing decisions. These insights can support targeted marketing campaigns, personalized offers, and improved customer engagement. In the future, the project can be improved by using real historical customer data, testing different machine learning algorithms such as Logistic Regression and Gradient Boosting, tuning model parameters, and evaluating performance on new customer data.